

Derivation of Diffusion Equations for High-Frequency Vibrations of Randomly Heterogeneous Structures

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Abstract

The evolution of the high-frequency vibrational energy density of slender visco-elastic structures such as Timoshenko beams or thick shells is depicted by transport equations, or radiative transfer equations in the presence of random heterogeneities. This paper deals with diffusion approximations of these equations and discuss the relevance of the Vibrational Conductivity Analogy invoked in the structural acoustics literature. Diffusive regimes are derived when random heterogeneities have correlation lengths either comparable to the wave length, or at an intermediate scale between the wave length and the medium size. Some numerical simulations are also shown.

Keywords: vibrations, high frequency, transport, radiative transfer, diffusion.

1 Introduction

The analysis of high-frequency vibrations of elastic structures has been the subject of intense research for the past decades either in the applied mathematics or the engineering mechanics literature. This issue has a great engineering relevance in the automotive or aerospace industries, for example, for the prediction of the vibro-acoustic behaviour of complex systems under broad-band external excitations. The best known and oldest technique is the method of geometrical optics, or WKB ansatz, for which there exists a vast literature: see for example [1, 2, 3] for some classical references in structural dynamics. This theory shows that the high-frequency energy density of a wave field is the solution of a transport equation. This result can also be obtained in the frame of the theory of multiple scattering of waves in random media, as done by Stott [4] or Weaver [5], among others, for electromagnetic or elastic media. It is also obtained in the frame of the (rigorous) mathematical theory of microlocal analysis developed in the late 1980's and early 1990's by several researchers [6, 7, 8, 9]

and extended to random media in [10, 11, 12].

In the limit of small transport mean free paths, transport equations can be approached by diffusion equations. This is the sense of the diffusion approximation originally used by Rybak [13] for elastic waves in a randomly heterogeneous plate. The author obtained a diffusion equation for the associated energy density, and this result was used later on by the Russian researchers whose papers are those cited by Nefske & Sung in their formulation of the Vibrational Conductivity Analogy (VCA) of structural dynamics [14]. But in this latter publication it appears that the conditions which have led to the diffusion approximation are no longer stated. Again, it was obtained in a random medium for multiply scattered waves – or can be obtained in a bounded homogeneous medium for multiply reflected waves: this is the diffuse field approximation used in room acoustics. For an homogeneous medium, the transport mean free path is infinite and a diffusive regime can theoretically never be reached. Despite its lack of theoretical ground, VCA has been the subject of several investigations in the structural acoustics literature, but the latter were always limited to simple homogeneous structures, typically rods, beams and more rarely plates; see for example [15, 16, 17, 18].

The purpose of this paper is to establish, from the theory of transport for high-frequency waves in randomly heterogeneous media [10, 11], the correct derivation of diffusion regimes in order to clarify the range of validity, if any, of the VCA. Two cases are considered: either (i) random heterogeneities at the same length scale as the wave length, or (ii) random heterogeneities at an intermediate scale between the scale of oscillations and the size of the medium. This more general theory of transport applies to heterogeneous media, does not neglect the possible interferences between energetic propagation modes, and does not rely on the artificial use of time and/or spatial averages, contrary to what is done in the VCA literature. In Section 2, some basic results pertaining to high-frequency wave propagation in an heterogeneous, visco-elastic medium are summarized. The transport or radiative transfer equations which characterize the energy density evolution are introduced. Section 3 deals with the derivation of diffusion equations from radiative transfer equations for case (i); the results presented in this paper extend the previous conclusions of [19, 20, 21] to heterogeneous background media and multi-mode diffusion. Section 4 deals with the derivation of the diffusive regime from stochastic transport equations in case (ii); some numerical simulations for a flat Mindlin plate are presented. Finally some conclusions are offered in Section 5.

2 Notations and hypotheses

In this section one summarizes the basic results obtained in [10, 11] for the transport properties of high-frequency waves in random media, and in [22] for visco-elastic materials with memory effect. For an heterogeneous, visco-elastic medium with statistically isotropic random perturbations of its mechanical parameters, such as the density and the Young or shear modulus, and embedded in the domain $\mathcal{O}$ of $\mathbb{R}^d$, the evolu-

tion of the high-frequency energy density is described by the following multigroup radiative transfer equations [10, 11, 22]:

$$\partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} = \sum_{\beta=1}^M \int_{S^{d-1}} \sigma_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') (w_\beta(\mathbf{x}, \mathbf{k}', t) - w_\alpha(\mathbf{x}, \mathbf{k}, t)) d\Omega(\hat{\mathbf{k}}') \quad (1)$$

which couple M different energy propagation modes (or rays). $\{f, g\} = \nabla_{\mathbf{k}} f \cdot \nabla_{\mathbf{x}} g - \nabla_{\mathbf{x}} f \cdot \nabla_{\mathbf{k}} g$ stands for the usual Poisson's bracket, S^{d-1} is the unit sphere of $\mathbb{R}^d$ and $d\Omega$ its Lebesgue measure, $\lambda_\alpha(\mathbf{x}, \mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ is the eigenfrequency associated to the mode α of which energy velocity is c_α , and $\sigma_{\alpha\beta} \geq 0$ is the scattering cross-section which gives the rate of conversion of an energy ray β in the direction $\hat{\mathbf{k}}'$ to another ray α in the direction $\hat{\mathbf{k}}$, at position $\mathbf{x}$ and wavenumber $|\mathbf{k}|$. In a deterministic medium all scattering cross-sections are zero and right-hand-sides vanish in (1) which become Liouville, or transport equations. The standard notation $\mathbf{k} = |\mathbf{k}|\hat{\mathbf{k}}$ is used throughout this paper. Note that the scalar Wigner measures $\{w_\alpha\}_{\alpha=1, \dots, N}$, hereafter called phase-space energy densities or specific intensities, have their values in $\mathbb{R}_+$. Then the space-time energy density of this medium is given by:

$$\mathcal{E}(\mathbf{x}, t) = \frac{1}{2} \sum_{\alpha=1}^M \int_{\mathbb{R}^d} w_\alpha(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} \quad (2)$$

and the space-time power flow density is given by:

$$\Pi(\mathbf{x}, t) = \frac{1}{2} \sum_{\alpha=1}^M c_\alpha(\mathbf{x}) \int_{\mathbb{R}^d} w_\alpha(\mathbf{x}, \mathbf{k}, t) \hat{\mathbf{k}} d\mathbf{k}. \quad (3)$$

These results are adapted to elastic waves in slender structures or porous media in [19, 20, 21], but they also apply to acoustic waves, electromagnetic waves, or the Schrödinger equation [10, 11, 12].

3 Diffusion regime in the limit of small scattering mean free paths

The scattering mean free path is defined for a particular propagating mode α by:

$$\ell_\alpha(\mathbf{x}, \mathbf{k}) = \frac{|\nabla_{\mathbf{k}} \lambda_\alpha|}{\Sigma_\alpha(\mathbf{x}, |\mathbf{k}|)}, \quad (4)$$

where Σ_α is the total scattering cross-section of a mode α defined by:

$$\Sigma_\alpha(\mathbf{x}, |\mathbf{k}|) = \sum_{\beta=1}^M \int_{S^{d-1}} \sigma_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') d\Omega(\hat{\mathbf{k}}'). \quad (5)$$

ℓ_α corresponds to the mean distance travelled by an energy ray in the mode α before its direction is significantly altered by scattering on the heterogeneities. Derivation of the radiative transfer equations (1) are carried out in the limit $\frac{\Lambda}{L} \rightarrow 0$, where Λ is a typical wavelength and $L \equiv c_\alpha t$ is a characteristic spatial size at which propagation is observed. The transport regime remains applicable as long as $\Lambda \ll \ell_\alpha$. A diffusion regime sets up when (i) the absorption of the medium is small, (ii) scattering is important, and (iii) one considers propagation distances much larger than the scattering mean free paths: $c_\alpha t \gg \ell_\alpha$. Therefore one introduces the small parameter $\epsilon = \frac{\ell_\alpha}{L}$ and let $\epsilon \rightarrow 0$ to obtain the diffusion limit by an asymptotic expansion method (the so-called Hilbert expansion method).

3.1 Properties of the collision operator

We first give the main properties of the so-called linear collision operator on the right hand side of (1). Define the average over the unit sphere S^{d-1} by:

$$\langle f \rangle = \frac{1}{\Omega_d} \int_{S^{d-1}} f(\hat{\mathbf{k}}) d\Omega(\hat{\mathbf{k}}), \quad \Omega_d = \int_{S^{d-1}} d\Omega(\hat{\mathbf{k}}),$$

and for $\mathbf{x} \in \mathcal{O} \subset \mathbb{R}^d$ and $|\mathbf{k}| > 0$ let:

$$\begin{aligned} \mathcal{C}(\mathbf{x}, |\mathbf{k}|)f(\hat{\mathbf{k}}) &= \tilde{\Sigma}(\mathbf{x}, |\mathbf{k}|)f(\hat{\mathbf{k}}) - \mathcal{K}(\mathbf{x}, |\mathbf{k}|)f(\hat{\mathbf{k}}), \\ \mathcal{K}(\mathbf{x}, |\mathbf{k}|)f(\hat{\mathbf{k}}) &= \frac{1}{\Omega_d} \int_{S^{d-1}} \tilde{\sigma}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') f(\hat{\mathbf{k}}') d\Omega(\hat{\mathbf{k}}'), \end{aligned}$$

where $\tilde{\Sigma} = \Omega_d^{-1} \Sigma \mathbf{C}^d$, $\tilde{\sigma} = \sigma \mathbf{C}^d$, $\mathbf{C} = \text{diag}(c_\alpha)_{\alpha=1, \dots, M}$, $\Sigma = \text{diag}(\Sigma_\alpha)_{\alpha=1, \dots, M}$, and σ is the $M \times M$ matrix of which elements are the scattering cross-sections $\sigma_{\alpha\beta}$ such that:

$$c_\beta^d \sigma_{\alpha\beta} \left(\mathbf{x}, \frac{c_\alpha}{c_\beta} |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' \right) = c_\alpha^d \sigma_{\beta\alpha}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}').$$

Then $\mathcal{C}(\mathbf{x}, |\mathbf{k}|)$ is self-adjoint on $H = (L^2(S^{d-1}))^M$; if we assume that $\tilde{\sigma}$ is such that:

$$\sup_{\mathbf{x} \in \mathcal{O}, |\mathbf{k}| > 0} \iint_{S^{d-1} \times S^{d-1}} \|\tilde{\sigma}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')\|^2 d\Omega(\hat{\mathbf{k}}) d\Omega(\hat{\mathbf{k}}') < +\infty$$

where $\|\tilde{\sigma}\|^2 = \sum_{\alpha, \beta=1}^M c_\beta^{2d} \sigma_{\alpha\beta}^2$, then $\mathcal{K}(\mathbf{x}, |\mathbf{k}|)$ is a Hilbert-Schmidt operator on H . Owing to the fact that the scattering cross-sections are non-negative functions, such that $\exists \alpha, \sigma_{\alpha\beta} > 0$ for each $\beta = 1, 2, \dots, M$ (otherwise the radiative transfer equations are split in sub-groups, and diffusion limits can be reached for each group separately, as in [19, 20, 21] for example), one can show in addition that the null space of $\mathcal{C}(\mathbf{x}, |\mathbf{k}|)$ is $N(\mathcal{C}(\mathbf{x}, |\mathbf{k}|)) = \{\phi \mathbf{c}; \phi = \langle \phi \rangle\}$ with $\mathbf{c} = \mathbf{C}^{-d} \mathbf{u}$, $\mathbf{u} = (1, 1, \dots, 1)$, and ϕ scalar functions independent of $\hat{\mathbf{k}}$. Indeed, $\mathcal{K}(\mathbf{x}, |\mathbf{k}|)$ is compact and strongly positive on the cone $K = \{f \in H; f_\alpha \geq 0, \alpha = 1, 2, \dots, M\}$, and one observes that $\mathbf{c}$ is a positive eigenvector of $\mathcal{K}(\mathbf{x}, |\mathbf{k}|)$; thus by the Krein-Rutman theorem (see for example [23]) it is the sole positive eigenvector associated to the simple eigenvalue $\rho(\mathcal{K})$, its spectral radius,

that is $N(\mathcal{C}(\mathbf{x}, |\mathbf{k}|))$ is the one-dimensional subspace of H spanned by $\mathbf{c}$. Now by Fredholm alternative, the equation $\mathcal{C}(\mathbf{x}, |\mathbf{k}|)f = g$ in $N(\mathcal{C}(\mathbf{x}, |\mathbf{k}|))^\perp = R(\mathcal{C}(\mathbf{x}, |\mathbf{k}|))$ has a unique solution f^* , which satisfies $(f^*, \phi\mathbf{c})_H = \phi\mathbf{c} \cdot \langle f^* \rangle = 0$ for any constant function ϕ .

3.2 Asymptotic analysis

First, it is convenient to rewrite the radiative transfer equations (1) for the phase-space energy densities $w_\alpha(\mathbf{x}, \mathbf{k}, t)$ in the form of transport equations for the specific intensities $a_\alpha(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}}, t) \equiv w_\alpha(\mathbf{x}, \mathbf{k}, t)$. This leads to the equations:

$$\partial_t a_\alpha + c_\alpha \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} a_\alpha - \nabla_{\mathbf{x}} c_\alpha \cdot (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} a_\alpha = \sum_{\beta=1}^M \int_{S^{d-1}} \sigma_{\alpha\beta} \left(\mathbf{x}, \frac{c_\alpha}{c_\beta} |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' \right) a_\beta \left(\mathbf{x}, \frac{c_\alpha}{c_\beta} |\mathbf{k}|, \hat{\mathbf{k}}', t \right) d\Omega(\hat{\mathbf{k}}') - \Sigma_\alpha a_\alpha, \quad (6)$$

$\alpha = 1, 2, \dots, M$, $\mathbf{I}_d$ being the identity matrix in $\mathbb{R}^d$. $(\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}}$ corresponds to the projection of the usual gradient $\nabla_{\mathbf{k}}$ defined in $\mathbb{R}^d$ onto its unit sphere S^{d-1} . In view of the conditions (i)–(iii), this system is rescaled as:

$$\epsilon \partial_t a_\alpha^\epsilon + c_\alpha \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} a_\alpha^\epsilon - \nabla_{\mathbf{x}} c_\alpha \cdot (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} a_\alpha^\epsilon + \epsilon \eta_\alpha a_\alpha^\epsilon = \epsilon^{-1} \sum_{\beta=1}^M \int_{S^{d-1}} \sigma_{\alpha\beta} \left(\mathbf{x}, \frac{c_\alpha}{c_\beta} |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' \right) a_\beta^\epsilon \left(\mathbf{x}, \frac{c_\alpha}{c_\beta} |\mathbf{k}|, \hat{\mathbf{k}}', t \right) d\Omega(\hat{\mathbf{k}}') - \epsilon^{-1} \Sigma_\alpha a_\alpha^\epsilon, \quad (7)$$

$\alpha = 1, 2, \dots, M$, where a small amount of dissipation $\epsilon \eta_\alpha(\mathbf{k})$ has been introduced. Let $\mathbf{a}^\epsilon = (a_1^\epsilon, a_2^\epsilon, \dots, a_M^\epsilon)$, then it is expanded in power series of ϵ as:

$$\mathbf{a}^\epsilon(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}}, t) = \mathbf{a}^{(0)}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}}, t) + \epsilon \mathbf{a}^{(1)}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}}, t) + \epsilon^2 \mathbf{a}^{(2)}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}}, t) + o(\epsilon^2)$$

and plugged into (7). Balancing order by order, one has to the leading order $O(\epsilon^0)$:

$$\mathcal{C}(\mathbf{x}, |\mathbf{k}|) \mathbf{C}^{-d} \mathbf{a}^{(0)} = \mathbf{0} \quad (8)$$

which shows that $\mathbf{a}^{(0)}$ does not depend on $\hat{\mathbf{k}}$, because $\mathbf{C}^{-d} \mathbf{a}^{(0)}$ lies in the null space of $\mathcal{C}(\mathbf{x}, |\mathbf{k}|)$. To the first order $O(\epsilon)$, one has:

$$-\Omega_d \mathcal{C}(\mathbf{x}, |\mathbf{k}|) \mathbf{C}^{-d} \mathbf{a}^{(1)} = (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}.$$

Its general solution is known to be:

$$\mathbf{a}^{(1)}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}}, t) = \mathbf{a}^{(11)}(\mathbf{x}, |\mathbf{k}|, t) - (\Lambda^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}(\mathbf{x}, |\mathbf{k}|, t) \quad (9)$$

where $\Lambda^*(\mathbf{x}, |\mathbf{k}|)$ is the solution of $\mathbf{C} \otimes \hat{\mathbf{k}} = \Omega_d \mathcal{C}(\mathbf{x}, |\mathbf{k}|) (\mathbf{C}^{-d} \Lambda^* \otimes \hat{\mathbf{k}})$, which is uniquely invertible since $\langle \mathbf{C} \otimes \hat{\mathbf{k}} \rangle = \mathbf{0}$. To the second order $O(\epsilon^2)$, we have:

$$\begin{aligned} -\Omega_d \mathcal{C}(\mathbf{x}, |\mathbf{k}|) \mathbf{C}^{-d} \mathbf{a}^{(2)} &= \partial_t \mathbf{a}^{(0)} + \boldsymbol{\eta} \mathbf{a}^{(0)} + (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(1)} \\ &\quad - (\mathbf{C} \otimes \nabla_{\mathbf{x}}) \left((\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes \mathbf{a}^{(1)} \right), \end{aligned}$$

with $\boldsymbol{\eta}(\mathbf{k}) = \text{diag}(\eta_\alpha)_{\alpha=1,\dots,M}$. The compatibility condition for $\mathbf{C}^{-d}\mathbf{a}^{(2)}$ to exist is that the right-hand-side in the above equation should belong to $N(\mathcal{C}(\mathbf{x}, |\mathbf{k}|))^\perp$, that is its averaged scalar product with any constant function $\phi \mathbf{c}$ should be zero. Therefore:

$$\mathbf{c} \cdot \left\langle \partial_t \mathbf{a}^{(0)} + (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(1)} - (\mathbf{C} \otimes \nabla_{\mathbf{x}}) \left((\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes \mathbf{a}^{(1)} \right) + \boldsymbol{\eta} \mathbf{a}^{(0)} \right\rangle = 0,$$

or, upon using (9) and reminding that neither $\mathbf{a}^{(0)}$ nor $\mathbf{a}^{(1)}$ depend on $\hat{\mathbf{k}}$:

$$\begin{aligned} & \mathbf{c} \cdot \partial_t \mathbf{a}^{(0)} + \mathbf{c} \cdot \langle (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(1)} \rangle - \mathbf{c} \cdot \left\langle (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes (\boldsymbol{\Lambda}^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)} \right\rangle \\ & + \mathbf{c} \cdot \left\langle (\mathbf{C} \otimes \nabla_{\mathbf{x}}) (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes (\boldsymbol{\Lambda}^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)} \right\rangle + \mathbf{c} \cdot \langle \boldsymbol{\eta} \rangle \mathbf{a}^{(0)} = 0. \end{aligned}$$

But by definition of the product of a third-order tensor and a second order tensor $(\mathbf{A} \otimes \mathbf{b})\mathbf{c} \otimes \mathbf{d} = (\mathbf{b} \cdot \mathbf{c})\mathbf{A}\mathbf{d}$, one first has:

$$\mathbf{c} \cdot \langle (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(1)} \rangle = \mathbf{c} \cdot \mathbf{C} \langle \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \rangle \mathbf{a}^{(1)} = 0$$

because $\langle \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \rangle = 0$; secondly:

$$\begin{aligned} \left\langle (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes (\boldsymbol{\Lambda}^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)} \right\rangle &= \mathbf{C} \langle \langle \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \rangle \boldsymbol{\Lambda}^* \langle \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \rangle \rangle \mathbf{a}^{(0)} \\ &= \frac{1}{d} \mathbf{C} \nabla_{\mathbf{x}} \cdot (\nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}) \boldsymbol{\Lambda}^* \end{aligned}$$

because $\langle \langle \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \rangle f(\hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}}) \rangle = \frac{1}{d} \nabla_{\mathbf{x}} \cdot f \nabla_{\mathbf{x}}$; finally:

$$\begin{aligned} & \left\langle (\mathbf{C} \otimes \nabla_{\mathbf{x}}) (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes (\boldsymbol{\Lambda}^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)} \right\rangle \\ &= (\mathbf{C} \otimes \nabla_{\mathbf{x}}) \left\langle (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes \boldsymbol{\Lambda}^* \langle \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \rangle \right\rangle \mathbf{a}^{(0)} \\ &= (\mathbf{C} \otimes \nabla_{\mathbf{x}}) (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) (\nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}) \boldsymbol{\Lambda}^* = \frac{d-1}{d} (\mathbf{C} \otimes \nabla_{\mathbf{x}}) (\nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}) \boldsymbol{\Lambda}^* \end{aligned}$$

since $\nabla_{\hat{\mathbf{k}}} \otimes \hat{\mathbf{k}} = \mathbf{I}_d$ and $\langle \mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}} \rangle = (1 - \frac{1}{d})\mathbf{I}_d$. Therefore $\mathbf{a}^{(0)}$ satisfies the diffusion equation:

$$\mathbf{c} \cdot (\partial_t \mathbf{a}^{(0)} - \nabla_{\mathbf{x}} \cdot (\nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}) \mathbf{D} + (\mathbf{C} \otimes \nabla_{\mathbf{x}}) (\nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}) \boldsymbol{\Lambda}^* + \langle \boldsymbol{\eta} \rangle \mathbf{a}^{(0)}) = 0 \quad (10)$$

with the diffusion matrix $\mathbf{D} = \frac{1}{d} \mathbf{C} \boldsymbol{\Lambda}^*$. This result extends the conclusions of [10, 19, 20, 21] to the case of heterogeneous media with multi-mode diffusion. The main advantage of the diffusion equation is that its solution is independent of the angular direction $\hat{\mathbf{k}}$. Its numerical solution is thererore much faster to construct than that of the full radiative transfer equation (1).

The "transmission relationship" or "energetic behaviour law" which links the power flow with the gradient of the energy density, as invoked in the VCA literature, is derived as follows. Taking the scalar product of equation (9) with $\mathbf{C}\mathbf{u}$, then pre-multiplying it with $\hat{\mathbf{k}}$ and integrating over S^{d-1} , yields the space-time power flow density vector as (see equation (3)):

$$\Pi^\epsilon(\mathbf{x}, |\mathbf{k}|, t) = -\epsilon \Omega_d (\nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}) \mathbf{D}(|\mathbf{k}|) \mathbf{u} + o(\epsilon). \quad (11)$$

Thus $\Pi^\epsilon(\mathbf{x}, |\mathbf{k}|, t) = 0$ in the limit $\epsilon \rightarrow 0$, as expected for a diffusive regime.

3.3 Examples

If $M = 1$, the diffusion equation (10) reduces to:

$$\partial_t a^{(0)} - \nabla_{\mathbf{x}} \cdot D \nabla_{\mathbf{x}} a^{(0)} + \ell^* \nabla_{\mathbf{x}} c \cdot \nabla_{\mathbf{x}} a^{(0)} + \langle \eta \rangle a^{(0)} = 0, \quad (12)$$

where the diffusion coefficient is $D(\mathbf{x}, |\mathbf{k}|) = \frac{1}{d} c(\mathbf{x}) \ell^*(\mathbf{x}, |\mathbf{k}|)$ and the transport mean free path ℓ^* is:

$$\ell^*(\mathbf{x}, |\mathbf{k}|) = c(\mathbf{x}) \left(\int_{S^{d-1}} \sigma(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') (1 - \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') d\Omega(\hat{\mathbf{k}}') \right)^{-1}. \quad (13)$$

The above equation is also applicable to the phase-space energy density $w^{(0)}$ or to the energy density in the limit $\epsilon \rightarrow 0$:

$$\mathcal{E}(\mathbf{x}, t) = \Omega_d \int_0^{+\infty} k^{d-1} a^{(0)}(\mathbf{x}, k, t) dk.$$

If the background medium is homogeneous, such that the celerities c_α are independent of $\mathbf{x}$ for $\alpha = 1, 2, \dots, M$, the diffusion equation reduces to:

$$\mathbf{c} \cdot (\partial_t \mathbf{a}^{(0)} - \nabla_{\mathbf{x}} \cdot \mathbf{D} (\nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)})^T + \langle \boldsymbol{\eta} \rangle \mathbf{a}^{(0)}) = 0.$$

Its solution also satisfies the equipartition rule $a_\alpha^{(0)}(\mathbf{x}, c_\alpha^{-1} |\mathbf{k}|, t) = a_\beta^{(0)}(\mathbf{x}, c_\beta^{-1} |\mathbf{k}|, t)$ because $\mathbf{a}^{(0)} = a^{(0)} \mathbf{u}$ by equation (8), and $a^{(0)}$ is a constant function of $\mathbf{k}$ which solves:

$$\partial_t a^{(0)} - \nabla_{\mathbf{x}} \cdot \tilde{D} \nabla_{\mathbf{x}} a^{(0)} + \tilde{\eta} a^{(0)} = 0 \quad (14)$$

with:

$$\tilde{D} = \frac{\sum_{\alpha, \beta=1}^M c_\alpha^{-d} D_{\alpha\beta}}{\sum_{\alpha=1}^M c_\alpha^{-d}}, \quad \tilde{\eta} = \frac{\sum_{\alpha=1}^M c_\alpha^{-d} \langle \eta_\alpha \rangle}{\sum_{\alpha=1}^M c_\alpha^{-d}},$$

as in [19, 20, 21].

4 Diffusion regime for the random Liouville equation

One considers now the case where specific intensities w_α , $\alpha = 1, 2, \dots, M$, are the solutions of stochastic Liouville equations:

$$\partial_t w_\alpha^\delta + \{\lambda_\alpha^\delta, w_\alpha^\delta\} = 0 \quad (15)$$

where $\lambda_\alpha^\delta(\mathbf{x}, \mathbf{k}) = c_\alpha^\delta(\mathbf{x}) |\mathbf{k}|$ and:

$$c_\alpha^\delta(\mathbf{x}) = c_\alpha^0 + \sqrt{\delta} c_\alpha^1 \left(\frac{\mathbf{x}}{\delta} \right). \quad (16)$$

$c_\alpha^0 > 0$ is a constant, $(c_\alpha^1(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$ is a second-order, mean-zero and mean-square stationary stochastic process indexed on $\mathbb{R}^d$ with values in $\mathbb{R}$. Its correlation function

is denoted by $\mathbf{x} \mapsto R_\alpha(\mathbf{x})$ such that $R_\alpha(\mathbf{x}) = \mathbb{E}\{c_\alpha^1(\mathbf{x} + \mathbf{y})c_\alpha^1(\mathbf{y})\}$, where $\mathbb{E}$ stands for the ensemble average. The correlation length δ of heterogeneities is an intermediate scale between the scale of oscillations and the size of the medium. Then it is proved rigorously in [24], for some light hypotheses on the random process $(c_\alpha^1(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$, that $\mathbb{E}\{w_\alpha^\delta\}$ converges weakly in $L^2(\mathbb{R}^d, d\mathbf{k})$ to $\bar{w}_\alpha$ as $\delta \rightarrow 0$, in the mean-square sense, that is:

$$\lim_{\delta \rightarrow 0} \mathbb{E} \left\{ \int_{\mathcal{O}} |(w_\alpha^\delta - \bar{w}_\alpha, f)_{L^2(\mathbb{R}^d, d\mathbf{k})}|^2 d\mathbf{x} \right\} = 0$$

for any test function $\mathbf{k} \mapsto f(\mathbf{k})$ in $L^2(\mathbb{R}^d)$. Furthermore, $\bar{w}_\alpha$ satisfies the Fokker-Planck equation:

$$\partial_t \bar{w}_\alpha + c_\alpha^0 \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \bar{w}_\alpha - \nabla_{\mathbf{k}} \cdot \mathbf{D}_\alpha(\mathbf{k}) \nabla_{\mathbf{k}} \bar{w}_\alpha = 0 \quad (17)$$

where the $d \times d$ diffusion matrix $\mathbf{D}_\alpha(\mathbf{k})$ is given by:

$$\mathbf{D}_\alpha(\mathbf{k}) = -\frac{1}{2} |\mathbf{k}|^2 \int_{\mathbb{R}} \nabla_{\mathbf{x}} \otimes \nabla_{\mathbf{x}} R_\alpha(c_\alpha^0 \hat{\mathbf{k}} u) du. \quad (18)$$

A formal derivation of the above diffusion equation is given in the subsequent section, while some numerical simulations are presented in section 4.2.

4.1 Formal derivation of the diffusion regime

Plugging in (16) into equation (15) yields:

$$\partial_t w^\delta + \left(c^0 + \delta^{\frac{1}{2}} c^1 \left(\frac{\mathbf{x}}{\delta} \right) \right) \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} w^\delta - \delta^{-\frac{1}{2}} |\mathbf{k}| \nabla_{\mathbf{x}} c^1 \left(\frac{\mathbf{x}}{\delta} \right) \cdot \nabla_{\mathbf{k}} w^\delta = 0,$$

dropping subscript α for clarity purposes. Then w^δ is expanded as:

$$w^\delta(\mathbf{x}, \mathbf{k}, t) = w^{(0)}(\mathbf{x}, \mathbf{y}, \mathbf{k}, t) + \delta^{\frac{1}{2}} w^{(\frac{1}{2})}(\mathbf{x}, \mathbf{y}, \mathbf{k}, t) + \delta w^{(1)}(\mathbf{x}, \mathbf{y}, \mathbf{k}, t) + o(\delta)$$

with $\mathbf{y} = \frac{\mathbf{x}}{\delta}$. The gradient may be expressed as $\nabla_{\mathbf{x}} \equiv \nabla_{\mathbf{x}} + \delta^{-1} \nabla_{\mathbf{y}}$, and to the leading order $O(\delta^{-1})$ one has:

$$c^0 \hat{\mathbf{k}} \cdot \nabla_{\mathbf{y}} w^{(0)} = 0, \quad \forall \hat{\mathbf{k}} \in S^{d-1}.$$

Thus $\nabla_{\mathbf{y}} w^{(0)} = \mathbf{0}$ and $w^{(0)}$ does not depend on $\mathbf{y}$. Balancing terms of order $O(\delta^{-\frac{1}{2}})$ leads to the equation:

$$c^0 \hat{\mathbf{k}} \cdot \nabla_{\mathbf{y}} w^{(\frac{1}{2})} - |\mathbf{k}| \nabla_{\mathbf{y}} c^1 \cdot \nabla_{\mathbf{k}} w^{(0)} = 0.$$

Its solution may be written $w^{(\frac{1}{2})}(\mathbf{x}, \mathbf{y}, \mathbf{k}, t) = \chi(\mathbf{y}, \mathbf{k}) \cdot \nabla_{\mathbf{k}} w^{(0)}(\mathbf{x}, \mathbf{k}, t)$ since, again, $w^{(0)}$ does not depend on $\mathbf{y}$; χ solves:

$$c^0 (\chi \otimes \nabla_{\mathbf{y}}) \hat{\mathbf{k}} - |\mathbf{k}| \nabla_{\mathbf{y}} c^1 = \mathbf{0},$$

that is $\chi(\mathbf{y}, \mathbf{k}) = \int_{\mathbb{R}_+} |\mathbf{k}| \nabla_{\mathbf{y}} c^1(\mathbf{y} - c^0 \hat{\mathbf{k}} u) du$ and:

$$w^{(\frac{1}{2})}(\mathbf{x}, \mathbf{y}, \mathbf{k}, t) = \left(\int_{\mathbb{R}_+} |\mathbf{k}| \nabla_{\mathbf{y}} c^1(\mathbf{y} - c^0 \hat{\mathbf{k}} u) du \right) \cdot \nabla_{\mathbf{k}} w^{(0)}(\mathbf{x}, \mathbf{k}, t). \quad (19)$$

The balance of $O(\delta^0)$ terms yields:

$$\partial_t w^{(0)} + c^0 \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} w^{(0)} + c^0 \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} w^{(1)} + c^1(\mathbf{y}) \hat{\mathbf{k}} \cdot \nabla_{\mathbf{y}} w^{(\frac{1}{2})} - |\mathbf{k}| \nabla_{\mathbf{y}} c^1 \cdot \nabla_{\mathbf{k}} w^{(\frac{1}{2})} = 0.$$

Taking the average of this equation with $\mathbb{E}\{w^{(1)}\} = 0$ and $\mathbb{E}\{w^{(0)}\} = \bar{w}^{(0)}$, one has:

$$\begin{aligned} \partial_t \bar{w}^{(0)} + c^0 \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \bar{w}^{(0)} &= \mathbb{E} \left\{ |\mathbf{k}| \nabla_{\mathbf{y}} c^1 \cdot \nabla_{\mathbf{k}} \left(|\mathbf{k}| \int_{\mathbb{R}_+} \nabla_{\mathbf{y}} c^1(\mathbf{y} - c^0 \hat{\mathbf{k}} u) du \cdot \nabla_{\mathbf{k}} w^{(0)} \right) \right\} \\ &\quad - \mathbb{E} \left\{ c^1(\mathbf{y}) \hat{\mathbf{k}} \cdot \nabla_{\mathbf{y}} \left(|\mathbf{k}| \int_{\mathbb{R}_+} \nabla_{\mathbf{y}} c^1(\mathbf{y} - c^0 \hat{\mathbf{k}} u) du \cdot \nabla_{\mathbf{k}} w^{(0)} \right) \right\}. \quad (20) \end{aligned}$$

But:

$$\begin{aligned} \mathbb{E} \{ |\mathbf{k}| \nabla_{\mathbf{y}} c^1 \cdot \nabla_{\mathbf{k}} (|\mathbf{k}| f(\mathbf{y}, \mathbf{k})) \} \\ = \mathbb{E} \{ \nabla_{\mathbf{k}} \cdot (|\mathbf{k}|^2 f(\mathbf{y}, \mathbf{k}) \nabla_{\mathbf{y}} c^1) \} - \mathbb{E} \{ f(\mathbf{y}, \mathbf{k}) \mathbf{k} \cdot \nabla_{\mathbf{y}} c^1 \}, \end{aligned}$$

and:

$$-\mathbb{E} \left\{ c^1(\mathbf{y}) \hat{\mathbf{k}} \cdot \nabla_{\mathbf{y}} (|\mathbf{k}| f(\mathbf{y}, \mathbf{k})) \right\} = \mathbb{E} \{ f(\mathbf{y}, \mathbf{k}) \mathbf{k} \cdot \nabla_{\mathbf{y}} c^1 \};$$

thus taking $f(\mathbf{y}, \mathbf{k}) \equiv |\mathbf{k}|^{-1} w^{(\frac{1}{2})}(\mathbf{x}, \mathbf{y}, \mathbf{k}, t)$ as given by equation (19), equation (20) is written:

$$\begin{aligned} \partial_t \bar{w}^{(0)} + c^0 \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \bar{w}^{(0)} \\ = \nabla_{\mathbf{k}} \cdot \left[|\mathbf{k}|^2 \mathbb{E} \left\{ \left(\int_{\mathbb{R}_+} \nabla_{\mathbf{y}} c^1(\mathbf{y} - c^0 \hat{\mathbf{k}} u) du \cdot \nabla_{\mathbf{k}} w^{(0)} \right) \nabla_{\mathbf{y}} c^1(\mathbf{y}) \right\} \right] \\ = \nabla_{\mathbf{k}} \cdot \left[|\mathbf{k}|^2 \left(\int_{\mathbb{R}_+} \mathbb{E} \left\{ \nabla_{\mathbf{y}} c^1(\mathbf{y}) \otimes \nabla_{\mathbf{y}} c^1(\mathbf{y} - c^0 \hat{\mathbf{k}} u) \right\} du \right) \nabla_{\mathbf{k}} \bar{w}^{(0)} \right]. \end{aligned}$$

The last equality is obtained by invoking the mixing lemmas 5.2 and 5.3 in [24] with some light assumptions introduced on the random process $(c^1(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$. The above equation is a diffusion equation of which diffusion matrix is given by:

$$\begin{aligned} \mathbf{D}(\mathbf{k}) &= |\mathbf{k}|^2 \int_{\mathbb{R}_+} \mathbb{E} \left\{ \nabla_{\mathbf{y}} c^1(\mathbf{y}) \otimes \nabla_{\mathbf{y}} c^1(\mathbf{y} - c^0 \hat{\mathbf{k}} u) \right\} du \\ &= -|\mathbf{k}|^2 \int_{\mathbb{R}_+} \nabla_{\mathbf{y}} \otimes \nabla_{\mathbf{y}} R(c^0 \hat{\mathbf{k}} u) du \\ &= -\frac{1}{2} |\mathbf{k}|^2 \int_{\mathbb{R}} \nabla_{\mathbf{y}} \otimes \nabla_{\mathbf{y}} R(c^0 \hat{\mathbf{k}} u) du \end{aligned}$$

because $\mathbf{y} \mapsto R(\mathbf{y})$ is an even function.

4.2 Numerical examples

The above results are applied to the case of bending or shear motion of a thick Mindlin plate ($d = 2$), for which $M = 3$ and $\alpha = \text{t, s or p}$ for either shear, axial bending (parallel to the wave vector $\mathbf{k}$) or transverse bending waves (parallel to the wave vector $\mathbf{l}$ such that $\mathbf{k} \cdot \mathbf{l} = 0$ on the mid-surface of the plate); see [19]. The correlation functions of random perturbations of the wave celerities c_α are chosen as:

$$R_\alpha(\mathbf{x}) = (\nu_\alpha c_\alpha^0)^2 e^{-(\gamma|\mathbf{x}|)^{2\xi}}, \quad \xi > \frac{1}{2}.$$

The diffusion matrix (18) is shown to be $\mathbf{D}_\alpha(\mathbf{k}) = \lambda_\alpha \mathbf{k} \otimes \mathbf{k} + \mu_\alpha |\mathbf{k}|^2 \mathbf{I}_2$ in this case, where:

$$\lambda_\alpha = 2\gamma\nu_\alpha^2 c_\alpha^0 \left[(\xi - 1)\Gamma\left(1 - \frac{1}{2\xi}\right) - \xi\Gamma\left(2 - \frac{1}{2\xi}\right) \right], \quad \mu_\alpha = \gamma\nu_\alpha^2 c_\alpha^0 \Gamma\left(1 - \frac{1}{2\xi}\right),$$

and $z \mapsto \Gamma(z) = \int_0^{+\infty} t^{z-1} e^{-t} dt$ stands for the usual Gamma function. Equation (17) is solved numerically by a direct Monte-Carlo method [25]. It consists in interpreting it as a Fokker-Planck equation for the marginal probability density function of an underlying diffusion process $U_\alpha(t) = (X_{\mathbf{x},\mathbf{k}}(t), K_{\mathbf{x},\mathbf{k}}(t), t \geq 0)$ defined on $\mathbb{R}^2 \times \mathbb{R}^2$, and which is the solution of Itô's stochastic differential equation:

$$dU_\alpha(t) = \mathbf{b}_\alpha(U_\alpha(t))dt + \boldsymbol{\sigma}_\alpha(U_\alpha(t))dB(t), \quad U_\alpha(0) = (\mathbf{x}, \mathbf{k}).$$

($B(t), t \geq 0$) is the usual brownian motion, and:

$$\mathbf{b}_\alpha(\mathbf{x}, \mathbf{k}) = \begin{pmatrix} -c_\alpha^0 \hat{\mathbf{k}} \\ (3\lambda_\alpha + 2\mu_\alpha)\mathbf{k} \end{pmatrix}, \quad \boldsymbol{\sigma}_\alpha(\mathbf{x}, \mathbf{k}) = \begin{bmatrix} \mathbf{0} & 0 & 0 \\ \mathbf{0} & \sqrt{\lambda_\alpha + \mu_\alpha} \mathbf{k} & \sqrt{\mu_\alpha} \mathbf{l} \end{bmatrix}.$$

Then $\bar{w}_\alpha$ with the initial condition $\bar{w}_\alpha(\mathbf{x}, \mathbf{k}, 0) = w_\alpha^0(\mathbf{x}, \mathbf{k})$ is obtained by:

$$\bar{w}_\alpha(\mathbf{x}, \mathbf{k}, t) = \mathbb{E}\{e^{-\eta_\alpha t} w_\alpha^0(X_{\mathbf{x},\mathbf{k}}(t), K_{\mathbf{x},\mathbf{k}}(t))\}$$

if some dissipation η_α is accounted for, that is, starting from a point $(\mathbf{x}, \mathbf{k})$, we simulate n paths of the process $U_\alpha(t)$ and average the values of the initial condition for these paths.

Our data for the simulations are as follows: $c_\alpha^0 = 1$ for the celerities, $\nu_\alpha = 0.5$ for their coefficients of variation, and $\xi = 1$ (the Gaussian model) or $\xi = 0.55$ (the exponential model corresponding to $\xi \rightarrow \frac{1}{2}$). The plate domain is $] -a, a[\times] -a, a[$ with $a = 10\gamma^{-1}$ and the initial condition is:

$$w_\alpha^0(\mathbf{x}, \mathbf{k}) = \delta(\mathbf{x}) \otimes \frac{1}{\sqrt{2}}(\cos \Theta - \sin \Theta, \cos \Theta + \sin \Theta)$$

where Θ is a uniformly distributed random variable within $]0, 2\pi[$; it gives the original direction of the wave vector, perpendicular to the initial wave front which can be seen on the figures below. Figure 1 and Figure 2 are the results obtained with $n = 4000$

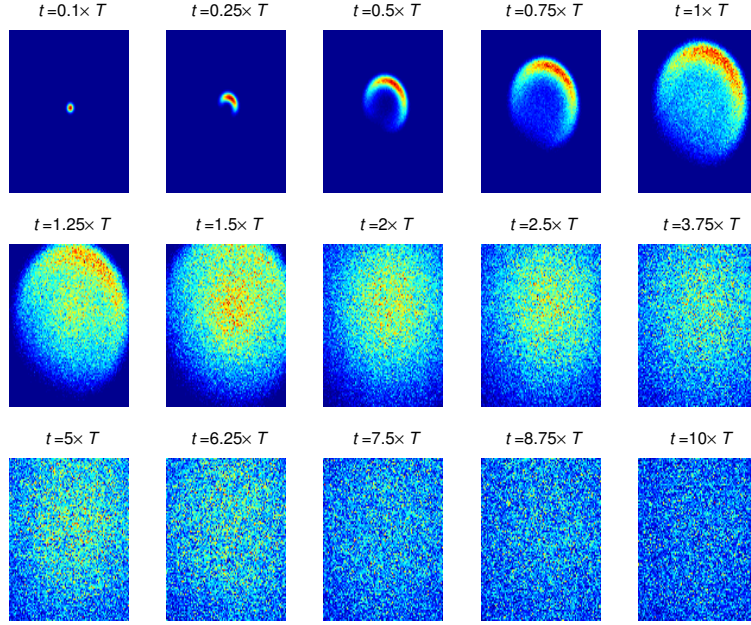


Figure 1: Evolution of the energy density in a flat Mindlin plate with Gaussian correlations of random perturbations of the wave celerities; $\gamma = 1$, $\nu_\alpha = 0.5$.

paths simulations for a Gaussian or an exponential model of correlation, respectively, $\gamma = 1$ for the correlation length scale of heterogeneities, and $t \in [0, 10T]$ with $T = (\gamma c_\alpha^0)^{-1}$. It is seen that in the Gaussian case, which corresponds to smooth random perturbations, the initial phase information is progressively lost as time evolves, up to an almost uniform distribution of energy over the plate. In the exponential case, which corresponds to rough random perturbations, the phase information is immediately lost and the energy remains concentrated around its source.

5 Conclusion

In this paper diffusion regimes which can be derived from the rigorous transport theory of high-frequency wave propagation have been presented. The objective of this study is to justify the Vibrational Conductivity Analogy of high-frequency structural vibrations invoked by some researchers. This approach is valid only when multiple scattering of the energy rays is possible; this is the case in the presence of random heterogeneities, as considered here, or boundaries and interfaces. Starting from radiative transfer equations, diffusion equations are derived in the limit of small scattering mean paths with respect to the structure size, or for long propagation times. The results presented in section 3 generalizes those of [19, 20, 21] to the case of multi-mode diffusion in heterogeneous background media. Starting from stochastic Liouville equations, Fokker-Planck equations are derived in the limit of a small correlation length of random perturbations of the eigenfrequencies of the energy modes. Classical diffu-

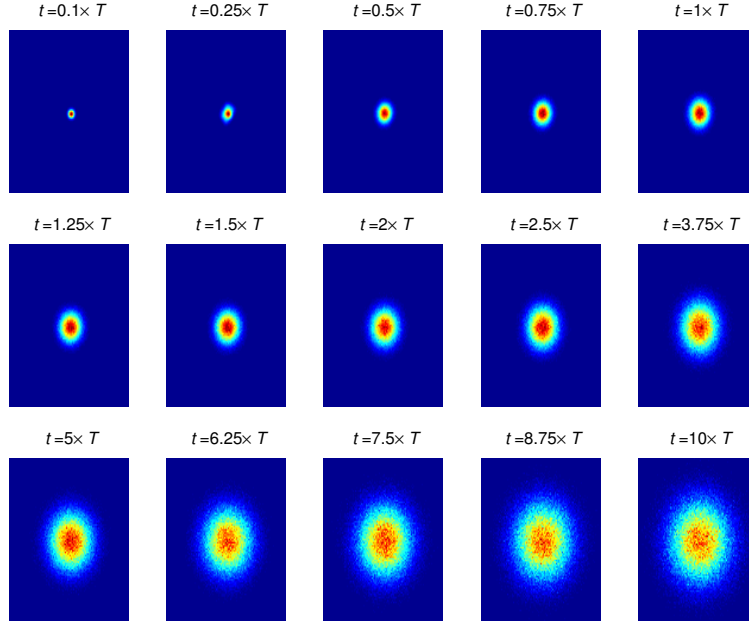


Figure 2: Evolution of the energy density in a flat Mindlin plate with exponential correlations of random perturbations of the wave celerities; $\gamma = 1$, $\nu_\alpha = 0.5$.

sion equations as those derived in Section 3 are easily recovered from Fokker-Planck equations in the limit, again, of small scattering mean free paths.

The purpose of this research is to construct a general model which depicts the energy density evolution at high-frequencies in complex structures, in order to predict, for example, the vibro-acoustic behaviour of such systems in the steady-state high-frequency regime, or their transient responses to impact loads such as pyrotechnic shocks. The main limitation of the proposed theory to date lies in the consideration of boundary and interface conditions for quadratic quantities (e.g., energy and power flow densities) consistent with the boundary/interface conditions imposed to the displacement and stress fields. Some preliminary results were obtained for the scalar wave equation with Dirichlet boundary conditions in a bounded smooth domain [26]; their generalization to polarized waves, such as elastic waves, and arbitrary boundary conditions is the subject of ongoing research.

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